One step at a time

Open the lesson from the PDF bookmarks.

Your teacher starts and advances the timer.

Use the Classroom Toolkit for Circuit Rescue, the logo-to-jingle companion and the NeuroArt finish check.

Neuro art and book checkout

I can join curved lines, round their corners, and follow our book-checkout routine.

See our plan

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch the line

Watch four curved lines cross.

Turn one sharp corner into a smooth curve.

Notice the large spaces left for color.

Draw and round

Write your name and class on the back.

Draw four to six crossing curves.

Round the sharp corners. Add dry color if time allows.

Choose your book

Cap markers and place artwork in the class tray.

When called, follow our library checkout routine.

Return with your book and read quietly.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Point to one corner you rounded and show that your table is ready.

Show your learning

Artwork in the class tray, or the Paint file saved where the teacher showed.

Canva logo 1: Start from a template

I can use a Canva logo template to make a logo and short slogan about something I am into.

Sign in

Get your device.

Open resources.svsd.net, then Canva.

Sign in with your school account.

Read tiny logos

Look at the real logos, shown very small.

Why can you still read them?

Tell a partner one reason.

Make your logo

Pick a logo template that fits one thing you are into.

Change the colors, the icon and the font.

Add a slogan of six words or fewer.

Small size check

Zoom out until your logo is stamp size.

Can you still read it?

Fix one thing.

Share with teacher

Choose Share.

Add your CIRC teacher's school account.

It is turned in when your teacher can open it.

Devices back

Close Canva.

Put your device back on the cart and plug it in.

Line up at the door.

Exit question

Answer at the line: what did the template do for you?

Could you do that yourself next time?

Show your learning

Share with your CIRC teacher's school account. It is turned in when your teacher can open it. Keep public and link sharing unchanged.

Logo to jingle: Give your brand a sound

I can turn my first logo into a short slogan and a 10-15 second musical message.

Open your first logo

Open resources.svsd.net, then Canva.

Find your first logo. Today you are adding sound.

Name the feeling or idea your logo should communicate.

Write the slogan

Write or choose three slogan ideas using six words or fewer.

Try an action plus a benefit: Build bright. Think big.

Choose one that matches your logo. Improve one word.

Plan your sound

Choose a mood: upbeat, calm, funny or dramatic.

Write your brand name, exact slogan and sound choice.

Share that plan with your teacher. You do not need an AI account.

Generate and combine

Watch the teacher turn a slogan into a jingle.

Copy your first logo onto a Canva video page. Add the slogan.

Use teacher-provided audio or an approved backing track. Aim for 10-15 seconds.

Generate and combine (continued)

Make the words easy to hear. Record the music source in your design.

Listen and fix

Play or perform your jingle for a partner.

Can your partner repeat the slogan and explain how it matches the logo?

Fix one unclear word, loud sound, or timing problem.

Share with teacher

Choose Share.

Add your CIRC teacher school account. It is turned in when your teacher can open it.

Save the logo, slogan, sound plan and any available audio.

Devices back

Return headphones and lower the volume.

Put your device back on the cart and plug it in.

Clear your table.

Show your learning

Share with your CIRC teacher's school account. It is turned in when your teacher can open it. Keep public and link sharing unchanged. Include the first logo, slogan, sound plan and available audio. A live performance counts.

Cardboard joins and careful cutting

I can make a stable cardboard connection and test it before adding decoration.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Watch the teacher cut away from the holding hand.

Compare a tab, slot, and taped join.

Watch the teacher test each join gently.

Make it work

Sketch a small shape that stands by itself.

Choose a tab, slot, or taped join.

Cut with approved scissors or use teacher-prepared pieces.
Assemble your shape.

Check one thing

Place your shape on the table and test its balance.

Change one weak join.

Compare the first and improved versions.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Show one stable connection and explain why it holds.

Show your learning

Three labeled joint samples and one stable shape in the class tray.

Make a package that protects

I can protect a model product, test the package, and improve one weak point.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Watch a model product move inside a loose package.

Notice padding, support, and an opening.

Watch one teacher-approved handling test.

Make it work

Choose the reusable model product provided.

Sketch a package that opens without damage.

Build using a material limit shown by your teacher.

Check one thing

Run the same gentle handling test twice.

Check movement, damage, and ease of opening.

Improve one weak point without adding unnecessary material.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Demonstrate how your package protects and opens, and explain one improvement.

Show your learning

Your package and one before and after change, shown to the teacher.

Paper Bridge

Transform paper into a bridge that spans a gap and carries increasing loads.

Get ready

Place the support blocks on the marks.

Count the load pieces.

Test safely

Keep hands below the table edge.

Place only approved loads on the bridge.

Plan the span

Choose a beam shape.

Predict where the paper may bend.

Sketch the bridge.

Build the bridge

Fold the paper into your chosen shape.

Tape only the planned joints.

Place the bridge across the fixed span.

Load the bridge

Add one load piece.

Record the supported total.

Repeat until the bridge fails.

Load the bridge (continued)

Change one weak part.

Reset the test

Return every load piece.

Recycle damaged paper.

Exit check

Write the highest supported load.

Name your useful change.

Show your learning

Your record sheet: most load held, one rebuild, same test repeated.

Crash Lander

Design a landing system that reduces impact on a fragile payload during a controlled drop.

Get ready

Count your allowed materials.

Place the payload beside the container.

Pick up one drop-zone team card.

Drop zone safety

Stay behind the drop-zone line.

Let the teacher release the lander.

Do the waiting job on your team card.

Drop zone safety (continued)

Move only when the teacher calls your team.

Plan the lander

Mark where the payload needs support.

Sketch one part that can bend safely.

Build the capsule

Place the payload inside the container.

Add only allowed materials.

Check that no part can fall off.

Test and revise

Predict the first failure point.

Watch the controlled drop.

Inspect the payload.

Test and revise (continued)

Change one feature.

Reset the drop

Return the reusable payload.

Collect every loose part.

Exit check

Draw the impact evidence.

Label your useful change.

Show your learning

Impact evidence and one revision.

Tank 1: Notice a useful problem

I can notice a real, manageable problem and explain who it affects.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Compare a broad wish with one specific everyday problem.

Notice who is affected and when it happens.

Make it work

Look at the approved classroom examples.

List or sketch three small problems.

Choose one that can be explored with classroom materials.

Check one thing

Tell a partner who has the problem.

Explain when it happens.

Narrow any idea that needs money, machinery, or changes outside our control.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Complete: Someone needs help with this problem when this happens.

Show your learning

Your problem sentence in the team folder: someone needs help with ___ when ___.

Cardboard Micro-Furniture

Build a miniature cardboard model of seating, storage, a work surface, or a quiet-space feature for a specific classroom user.

Get ready

Choose one user card.

Choose one classroom location.

Cut safely

Cut only on the table.

Keep fingers outside the cutting path.

Carry scissors closed.

Plan for the user

Draw the furniture feature.

Label how the user reaches it.

Mark the model size.

Build the model

Cut the planned cardboard shapes.

Use two connection methods.

Keep the model table sized.

Test the user path

Move the paper user toward the model.

Check the reach point.

Check the model stability.

Remove a problem

Change one blocked or weak feature.

Repeat the user path.

Sort the materials

Close the scissors.

Return usable cardboard.

Store the model flat.

Exit check

Point to the user feature.

Explain the problem you changed.

Show your learning

A mini model tested with a paper user figure.

Tank 2: Listen before choosing

I can ask a useful question and use the answer to improve my problem statement.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Compare a leading question with an open question.

Watch a short respectful interview.

Make it work

Prepare two questions about the task or object.

Ask an approved partner, or use the teacher's user card.

Record the useful answer with a short note or drawing.

Check one thing

Compare the answer with your first idea.

Change the problem statement if needed.

Ask whether you are describing a need or jumping to a solution.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Share one thing you learned from the user and what it changed.

Show your learning

A problem statement changed by one user's answer, in the team folder.

Catch up and finish

I can find my missing work, finish it, and show it is done.

Check your list

Look at the board.

Find your missing piece of work.

Tell your partner

Tell a partner what you will finish today.

Pick your job

Finish, redo a test, or help a teammate.

Tell the teacher your pick.

Catch up

Work on your one job.

Ask for help early.

Check it

Show your finished piece to a partner or the teacher.

Reset together

Return tools and materials.

Put finished work in the right tray.

Clear your table.

Show your result

Say what you finished today.

Show your learning

The missing piece of work, shown to the teacher.

Plan a small casting without pouring

I can draw a useful small object, estimate its size, and compare possible materials.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Look at a dry mold or its measured drawing.

Identify the inside space that shapes the object.

Compare a cardboard model with a teacher-cleared cured example or picture.

Make it work

Sketch a small token, tag, or game piece.

Measure a rectangular practice cavity or its drawing.

Use length, width, and depth to estimate its volume.

Check one thing

Compare making your design from cardboard and casting material.

Name one reason for each choice.

Improve the design using the teacher's size limit.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Show a labeled design and explain why your chosen material fits its use.

Show your learning

Your labeled drawing and size estimate.

KidWind 1: Make the wind do work

I can build blades that turn wind into work and measure how far they lift a load.

Get ready

Sit with your team.

Read the goal: I can make the wind do work.

Tell your partner

Tell your partner one place you feel strong wind.

Station safety

Watch the cup rise from the start line.

Stay out of the spin path.

Only the teacher turns the fan on or off.

Build a rotor set

Cut three blades from cardstock.

Tape each blade to a blade support.

Set each blade at the same angle as the example.

Test at the station

Your team gets one 10 second trial.

Watch the cup rise.

Record the centimeters lifted.

Reset the station

Fan off first.

Rotor sets go flat in your team folder.

Cup and load pieces go back in the station tray.

Show your result

Say how many centimeters your blades lifted the cup.

Name one thing you might change.

Show your learning

Your team's centimeters lifted in 10 seconds, on your record sheet and the class chart.

Tank 3: Try three ideas

I can sketch different solutions and choose one using a clear reason.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

See three different ways to solve one small problem.

Compare usefulness, simplicity, and available materials.

Make it work

Sketch three different solutions.

Label the part that does the work.

Circle one idea you can test with classroom materials.

Check one thing

Show the ideas to a partner.

Name one strength and one concern for the chosen idea.

Revise the sketch before building.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Show three ideas and explain why one is the best first test.

Show your learning

Three sketches, one chosen idea and the reason.

KidWind 2: Change one thing

I can change one thing about my blades and keep everything else the same.

Get ready

Find your team folder and your Set A rotor.

Read the goal: change one thing only.

Read the chart

Look at last time's class chart.

Watch a fair test

Watch the teacher change only the blade angle.

Notice what stays the same.

Build Set B

Pick one thing to change: number, size, angle or shape.

Keep everything else the same as Set A.

Write the one change on your record sheet.

Test A and B

Test Set A once.

Test Set B once.

Record centimeters lifted for each trial.

Reset the station

Fan off first.

Rotor sets go flat in your team folder.

Cup and load pieces go back in the station tray.

Share

Tell the class which set lifted the cup higher and by how many centimeters.

Show your learning

Set A and Set B results in centimeters, with the one change named. These are first results.

Tank 4: Plan the first build

I can make a labeled plan and choose a small, testable first version.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Watch a sketch become a parts list.

Separate essential working parts from decoration.

Make it work

Draw the first version from a useful view.

Label its working parts and approximate sizes.

Choose materials and agree on the first build jobs.

Check one thing

Explain how a user will try the model.

Remove any part that does not help the first test.

Show the teacher your plan before collecting supplies.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Show a labeled plan, short materials list, and one thing your model must do.

Show your learning

A labeled plan with a materials list.

KidWind 3: Build the best set

I can use class evidence to build a better set of blades.

Get ready

Sit with your team.

Read the goal: use evidence to build better.

Find your records

Find your Set A and Set B results.

Read the class chart

Look for the sets that lifted the cup highest.

What did those blades have in common?

Build your best set

Use the evidence to design one best set.

Say which chart result supports your choice.

Repeat fair trials

Run two or three 10-second trials, as your teacher directs.

Record every result in centimeters, even the low ones.

Reset the station

Fan off first.

Rotor sets go flat in your team folder.

Cup and load pieces go back in the station tray.

Compare

Compare your best set with your first set, in centimeters.

Show your learning

Your best set's results from two or three 10-second trials, as your teacher directs, in centimeters and compared with your first set.

Tank 5: Practice the tricky part

I can practice one connection or moving part before using it in my prototype.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Watch a tab, hinge, slider, or string pull.

Identify where friction, looseness, or weakness could occur.

Make it work

Choose the hardest working part in your plan.

Build a small practice sample using scrap.

Try the movement or connection several times.

Check one thing

Name the first weak or unreliable point.

Change one feature.

Keep the stronger sample as a reference.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Demonstrate your practice part and name the improvement.

Show your learning

An improved practice sample in the class crate.

KidWind 4: Tell what the wind taught us

I can explain what we changed, what happened, and the proof.

Get ready

Quick recap: what did our blades do?

Find your records

Get your folder and your best set.

Final test

Run your best set for one 10 second trial.

Record the centimeters lifted.

Plan your talk

Plan a 30 to 45 second talk.

Say what we changed, what happened, and our proof in centimeters.

Share your talk

Give your talk to another team.

Listen and ask one question.

Reset the station

Fan off first.

Rotor sets go flat in your team folder.

Cup and load pieces go back in the station tray.

Show your result

Say one thing the wind taught you.

Show your learning

A 30 to 45 second talk with a centimeter result as proof.

Tank 6: Build version one

I can build a rough model that shows my idea's main job.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Watch the main working part being built first.

Notice that a rough prototype can answer a useful question.

Make it work

Build the essential working part from your plan.

Add only the support needed to try it.

Keep a sketch or picture of version one using the approved classroom method.

Check one thing

Ask a partner to identify the model's job.

Try the intended action once.

Mark one feature that needs a test next time.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Show version one doing its main job, or identify the exact part that needs repair.

Show your learning

Version one in your crate cell with its folder.

Human Program

Create and debug an exact sequence of commands that guides a human robot through a task.

Get ready

Choose one short task card.

Place the grid mat on the floor.

Robot safety

Move only at walking speed.

Stop at every obstacle.

Stop when anyone says stop.

Write commands

Write one command per line.

Use only approved command words.

Run the program

Give the commands to the human robot.

Watch each command run exactly.

Mark the first wrong result.

Debug one line

Rewrite the first wrong command.

Run the complete program again.

Switch roles

Choose a new human robot.

Run the corrected program once.

Reset the grid

Stack the command cards.

Roll the grid mat.

Exit check

Copy the corrected command.

Name the result it changed.

Show your learning

The first bug and the command change that fixed it.

Tank 7: Plan a fair test

I can plan a repeatable test that checks what my design claims.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Compare two tests that change several things.

Watch a fair test keep the object, starting point, and action consistent.

Make it work

Write or draw what the product claims to do.

Choose one thing to count, measure, or observe.

Plan three trials with the same starting conditions.

Check one thing

Walk through the test without rushing.

Ask a partner what must stay the same.

Get the teacher's approval before the real trials.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Explain your test, what stays the same, and what result would count as success.

Show your learning

A written test plan: what changes and what stays the same.

Cargo Sorter

Design a mechanical sorting system that routes model cargo by one measurable property.

Get ready

Place mixed cargo in the starting cup.

Choose two ending bins.

Sorter safety

Keep fingers outside every moving path.

Keep magnets away from electronics.

Feed one cargo item at a time.

Plan the sort

Choose one sorting rule.

Draw the travel path.

Label both ending bins.

Build the sorter

Build the starting ramp.

Add one part that sends pieces to a bin.

Place the bins under the exits.

Run the batch

Feed one mixed item.

Record its ending bin.

Repeat for all test pieces.

Fix the errors

Change the sorting part.

Test all pieces again.

Reset the cargo

Count every cargo item.

Return magnets to the marked box.

Exit check

Write the final error count.

Name the sorting rule.

Show your learning

Batch accuracy and the fix that stopped the most common error.

Tank 8: Test and record

I can run the same test three times and record what actually happens.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Watch one trial from the same starting position.

See a record include an unsuccessful result honestly.

Make it work

Reset the test to the agreed starting position.

Run three trials and record each result.

Keep the design unchanged until the trials are finished.

Check one thing

Compare all three results.

Point to the most useful pattern or failure.

Choose one problem to fix next.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Show three results and identify one specific design problem.

Show your learning

Three recorded trials in the team folder.

Helping Hand

Build a hand-operated tool that safely grips and moves an object beyond normal reach.

Get ready

Read one role card.

Join the one shared educational prototype team.

Build safely

Cut only on the table.

Keep fingers outside the cutting path.

Carry scissors closed.

Build safely (continued)

Keep the prototype away from bodies.

Plan the grip

Choose one lightweight test object.

Sketch the handle.

Sketch the moving connection.

Plan the grip (continued)

Mark the soft part that holds the object.

Build one shared part

Do only the work on your role card.

Put your finished work in the team tray.

Show it to the build lead.

Test on the stand

The test team places the build on the test stand.

The test team moves one approved object.

The record team writes the result with no student name.

Improve control

The fix team changes one moving or soft part.

The test team repeats the stand test.

Reset the build

Close the scissors.

Store the shared prototype as one item.

Exit check

Name your role contribution.

Cite one stand test result.

Show your learning

One successful move and how it serves the need.

Tank 9: Improve one feature

I can use test evidence to change one feature and check whether it helps.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Compare a design change with a decorative change.

Trace one failed test to the part causing it.

Make it work

Choose one problem from your test record.

Change the relevant shape, connection, or movement.

Repeat the same three-trial test.

Check one thing

Compare the old and new results.

Keep the improvement or explain why you will undo it.

Record what the evidence suggests next.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Show one change you made after testing. Compare the results before and after.

Show your learning

Before and after results for one change.

Upcycle Lab

Transform a clean discarded item into a useful prototype with a clearly defined user and function.

Get ready

Choose one inspected reclaimed object.

Place it on the project mat.

Material safety

Use only the approved clean object.

Cut only on the table.

Keep fingers outside the cutting path.

Material safety (continued)

Carry scissors closed.

Choose a new use

List two useful object properties.

Name one user.

Write one new function.

Build the new use

Sketch the planned change.

Cut only approved added material.

Build the new function.

Test the function

Run one clear use test.

Record the result.

Find one weak feature.

Improve the use

Change the weak feature.

Repeat the use test.

Prepare a before-and-after explanation.

Reset the lab

Close the scissors.

Sort every unused material.

Exit check

Name the reused property.

State the new function.

Show your learning

Your object's new job, explained before and after.

Tank 10: Let someone try

I can watch a user try my idea and learn without coaching every move.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Watch a user receive one short instruction.

Notice how the designer listens without taking over.

Make it work

Tell an approved partner the product's job.

Let the partner try it using the approved test.

Record one clear action, one confusion, and one useful comment.

Check one thing

Ask what was easiest and hardest.

Choose one change that would help the user.

Try a quick paper or scrap version of that change.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Explain one user difficulty and the change you will make.

Show your learning

One user difficulty and the change you plan.

Accessibility Design Sprint

Design and test a small prototype that reduces a specific barrier involving reach, grip, vision, hearing, movement, or attention.

Get ready

Choose one provided user card.

Read the user goal aloud.

Design safely

Use only the facts on the user card.

Cut only on the table.

Keep fingers outside the cutting path.

Design safely (continued)

Carry scissors closed.

Name the problem

Name the tool or space problem.

Sketch three possible changes.

Circle one idea.

Build one feature

Cut only the planned shapes.

Build the smallest useful feature.

Add a large clear label.

Run the card test

Use the provided test card.

Observe one access task.

Record where the task stops.

Fix the problem

Change one blocked feature.

Repeat the same card test.

Reset the sprint

Close the scissors.

Store the prototype with its card.

Exit check

Write what became easier.

Write what remains difficult.

Show your learning

A 30 second pitch: user goal, barrier, test, result, change.

Tank 11: Use materials wisely

I can choose materials that do the job while reducing unnecessary waste.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Compare two materials doing the same small job.

Notice what can be reused or made smaller.

Make it work

Sort the design's parts into essential and optional.

Compare two suitable materials for one part.

Improve that part without making the product harder to use.

Check one thing

Check that the main job still works.

Count pieces or compare material size before and after.

Sort useful scraps for another group.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Show a useful material choice and explain what you saved or improved.

Show your learning

One part made smaller or reused, noted in the folder.

Moving Picture Machine

Create a sequence device that uses small image changes to produce the illusion of motion.

Get ready

Choose flip, wheel, or strip motion.

Place the frame paper in order.

Fastener safety

Keep the device at table level.

Use fasteners only through marked holes.

Ask the teacher about a sharp fastener.

Plan the motion

Draw a clear start pose.

Draw a clear finish pose.

Mark the matching dot.

Draw the frames

Change one small part per frame.

Keep each drawing on the registration mark.

Number the frames.

Assemble the motion

Stack or attach the frames in order.

Secure the approved fastener.

Move the device slowly.

Smooth the motion

Find the largest visual jump.

Revise one frame.

Test at two speeds.

Sort the pieces

Return unused fasteners.

Clip your frames together.

Exit check

Show the finished motion once.

Name the frame you changed.

Show your learning

A running animation with one smoothed frame.

Tank 12: Name and package

I can communicate my product's purpose with a clear name, logo, and simple package.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Compare a readable product label with a crowded one.

Watch a package protect the model and open easily.

Make it work

Choose a clear product name and reuse your logo skills.

Make a label that tells the product's job.

Create or improve a simple protective package.

Check one thing

Ask a partner to read the label without help.

Check opening, fit, and the approved handling test.

Revise one confusing or wasteful detail.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Show a readable product label and a package that opens and protects.

Show your learning

A readable label and a protective package.

Fix, Remix, or Invent Studio

Choose to repair a flawed prototype, remix an earlier challenge with a new constraint, or invent a small solution from a prompt, then test it with evidence.

Get ready

Choose fix, remix, or invent.

Read the matching safety card.

Studio safety

Inspect every material before use.

Cut only on the table.

Keep fingers outside the cutting path.

Studio safety (continued)

Carry scissors closed.

Stop any unexpected hazard.

Set the test

Write the prompt.

Write the rule you must follow.

Define one success test.

Run the first test

Build or inspect the first version.

Run the success test.

Record the most useful evidence.

Make one change

Choose one change because of the test result.

Cut only the planned material.

Complete the change.

Retest the change

Repeat the same success test.

Keep, revise, or reject the change.

Record your reason.

Reset the studio

Close the scissors.

Sort all reusable parts.

Exit check

Compare the two test results.

Name your next change.

Show your learning

Your chosen path, a baseline test and a first attempt.

Tank 13: Explain your idea

I can explain the problem, my solution, and evidence that helped me improve it.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Hear a short pitch with a problem, idea, proof, and next step.

Notice a speaker showing the product while explaining it.

Make it work

Make four short cue cards or picture prompts.

Choose one clear test result as proof.

Prepare a simple demonstration of the product.

Check one thing

Tell the story to one partner.

Ask the partner to repeat the main idea.

Cut details that hide the problem, solution, or proof.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Give a short practice pitch that includes the problem, solution, and one piece of evidence.

Show your learning

Four cue cards and one practice pitch.

Innovation Studio 2: Change one thing and retest

I can change one part of my design, run the same test, and decide to keep or undo it.

Get ready

Get your model and record sheet.

Read the goal: one change, same test.

Find your baseline

Find last time's test result.

Read your baseline

Look at last time's test result.

Pick the one part that held you back.

Make one change

Change one part only.

Keep the test the same.

Retest

Run the same test again.

Record the new result next to the old one.

Keep or undo

Did the change help?

Keep it or undo it. Write why.

Reset together

Return tools and reusable pieces.

Put unfinished work in the class tray.

Clear your table and floor.

Show your result

Show your before and after results to the teacher.

Show your learning

Before and after results and a keep or undo reason.

Tank 14: Rehearse and refine

I can practice a clear presentation and improve it after specific feedback.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Watch how speakers take turns and show their model on the table.

Compare useful feedback with a vague compliment.

Make it work

Choose who will explain or demonstrate each part.

Rehearse once while a partner times the explanation.

Practice one likely question and an honest answer.

Check one thing

Ask for one strength and one specific improvement.

Change the unclear part.

Rehearse again with a reliable reset.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Deliver your revised pitch and show how the product resets for the next visitor.

Show your learning

A revised, timed pitch.

Innovation Share

I can show my problem, process, evidence and next step to my classmates.

Get ready

Get your model and folder.

Read the goal: show four parts.

Find your work

Find one record from KidWind or a challenge.

Watch an exhibit

See the four parts: problem, process, evidence, next step.

Build your exhibit

Set out your model.

Add one record as evidence.

Write four short labels.

Gallery walk

Visit two exhibits.

Leave one kind, specific comment.

Take home decision

Decide: take home, class sample, or recycle.

Empty your crate cell.

Reflect

Name one thing you would change next time.

Show your learning

A four-label exhibit, then a take-home decision.

Tank 15: Share and reflect

I can show my design, listen to feedback, and explain how it changed.

Get ready

Read today's goal.

Look at the example.

Tell your partner

Tell your partner one useful detail.

Watch and try

Watch a visitor approach, hear a short pitch, and ask one question.

Notice how the presenter resets the space.

Make it work

Set out the prototype, label, and one piece of test evidence.

Take turns presenting and visiting other teams.

Give one useful observation and one respectful question.

Check one thing

Compare your first sketch with your final version.

Choose the most useful change.

Name one thing you would test next.

Reset together

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Show your result

Explain how evidence changed your design and identify a sensible next test.

Show your learning

Your first sketch next to your final model, with evidence. Then decide what goes home.

Hydroponics Flow Lab

Model a gravity-fed hydroponic path that distributes water evenly without leaking or flooding.

Get ready

Place the dry tray on the wet mat.

Measure the test water.

Wet tool safety

Keep water inside the tray.

Wipe each spill with a towel.

Do not cut tubing yourself.

Wet tool safety (continued)

Keep electronics away from the wet zone.

Plan the flow

Draw the water cup.

Draw the water path.

Mark each plant spot and the end cup.

Build the dry model

Mark the cut line on the tube.

Give the tube to the teacher.

Wait for the cut, then connect the tube path.

Build the dry model (continued)

Check every connection while dry.

Run the water

Pour the measured water into the water cup.

Watch each plant spot.

Measure the collected water.

Change the flow

Change one slope, hole, or loose connection.

Repeat with the same water amount.

Record the new collected amount.

Dry the lab

Drain the tray into the sink bucket.

Place wet parts on the drying tray.

Dry the table and floor.

Exit check

Write the collected amount.

Name the flow change.

Show your learning

A measured flow result and one change that made the flow more even.

Tinkercad 1: Move, turn, size

I can place shapes in Tinkercad and control their size, height and turn.

Sign in

Get your laptop.

Join the class with the code and your nickname.

Tell your partner

Tell your partner something you have seen that was 3D printed.

Watch the handles

Watch the teacher drag a box.

See how to size, lift and turn it.

Make a name plate

Drag a box for the base.

Add letters with your initials or nickname.

Size the letters to fit the base.

Check it

Look from the side.

Letters sit on the base, not floating or sunk.

Show a partner your side view on screen.

Save and shut

Check that your work saved.

Close the lid with two hands.

Put the laptop back in its slot and plug it in.

Tell how

Tell a partner how you kept the letters on the base.

Show your learning

A name plate with letters sitting on the base, shown on screen.

Tinkercad 2: Holes and group

I can use a hole shape and group to make a keychain tag with a working hole.

Sign in

Get your laptop.

Open Tinkercad with the class code.

Recap

Show a partner how to lift a shape.

Watch a hole

Watch a cylinder turn into a hole.

Watch Group cut it out.

Make a tag

Make a box for your keychain tag.

Set a cylinder to Hole and push it through a corner.

Select both and choose Group.

Check the hole

Can you see through the hole?

If not, push it all the way through.

Show a partner the hole from underneath.

Save and shut

Check that your work saved.

Close the lid with two hands.

Put the laptop back in its slot and plug it in.

Tell how

Tell a partner how you made the hole go all the way through.

Show your learning

A keychain tag with a hole you can see through.

Tinkercad 3: Align and mirror

I can use duplicate, mirror and align to make both sides of a model match.

Sign in

Get your laptop.

Open Tinkercad with the class code.

Recap

Tell a partner what Group does.

Watch mirror

Watch one half become two with Mirror.

Watch Align line them up.

Build a matching model

Build one half of a trophy or rocket.

Duplicate it, then Mirror it.

Use Align to line both halves up.

Check both sides

Do both sides match?

Fix one side if they do not.

Show a partner both sides on screen.

Save and shut

Check that your work saved.

Close the lid with two hands.

Put the laptop back in its slot and plug it in.

Name the tool

Name the tool that helped your two sides match.

Show your learning

A model with matching sides.

Tinkercad 4: Design a skate park piece

I can design a skate park piece that fits the size rule and has a flat bottom.

Sign in

Get your laptop.

Open Tinkercad with the class code.

Measure a board

Hold a test board. How wide is it?

Watch the size rule

No piece bigger than 80 by 60 by 40 millimeters.

Flat bottom, at least 40 millimeters wide.

Design your piece

Choose a kicker ramp, quarter pipe, grind box or rail.

Start from a wedge or a box.

Check your size with the numbers.

Teacher check

Show the flat bottom and the size.

Fix anything flagged.

Save and shut

Check that your work saved.

Close the lid with two hands.

Put the laptop back in its slot and plug it in.

Show your result

Tell a partner which piece you made and its size.

Show your learning

A skate park piece that passes the size and flat bottom check.

Tinkercad 5: Fix, export, queue

I can fix my design, export it, and put it in the class print folder with a clear name.

Sign in

Get your laptop.

Open your skate park piece.

Read your note

Read the teacher note about your design.

Fix flagged parts

Fix what the note says.

Check the size and flat bottom again.

Export

Choose Export, then STL.

Name the file with your table number and piece.

Queue it

Put your file in the class print folder.

Check that your file name is there.

Remember your file label and folder.

Save and shut

Check that your work saved.

Close the lid with two hands.

Put the laptop back in its slot and plug it in.

Tell the teacher

Tell the teacher your file label and where you saved it.

Show your learning

An STL file in the class print folder, named with your table number.

Tinkercad 6: Build day and pitch

I can assemble our table's skate park, test it, and pitch what we made.

Get ready

Find your table's pieces.

Read the goal: build, test, pitch.

Plan the park

Decide where each piece goes.

Build the park

Place your table's pieces on the base sheet.

Use cardboard for any piece not printed.

Test with a board

Roll a board over each piece.

Fix one thing that catches.

Pitch

Give your pitch.

Say what it is, what was hard, and what you would change.

Take home rule

Follow the take home rule for printed pieces.

Recycle cardboard scraps.

Show your result

Name one skill you used from all six visits.

Show your learning

Your pitch, and your piece tested on the table park.

Outdoor Microclimate Map

Compare conditions across outdoor classroom zones and recommend where different learning activities should happen.

Get ready

Clip the zone map to your board.

Check your assigned zones.

Walk out

Walk with your weather team.

Use tools safely

Carry the thermometer by its case.

Leave unknown plants untouched.

Stay inside each marked zone.

Observe two zones

Mark sun or shade.

Mark the surface condition.

Watch the wind ribbon.

Take readings

Take one approved reading in each zone.

Write each number on the map.

Build the map

Add both zones to the class key.

Choose the best zone for one activity.

Pack the tools

Return each tool to its case.

Clip all maps together.

Return inside

Walk back with your weather team.

Exit check

Write two observations that support your zone choice.

Show your learning

Two mapped zones and one activity recommendation.

Reflect, reset, take home

I can look back at my year of work, name what I learned, and take my work home.

Get ready

Find your work from this year.

Read the goal: look back and take it home.

Tell your partner

Tell a partner your favorite CIRC day.

Lay out your year

Lay out your records, models and files.

Find one piece you are proud of.

Write your reflection

Draw or write one thing you learned.

Write one thing you would change.

Share one thing

Share your proud piece with a partner.

Take it home

Pack your work to take home.

Help count the reusable sets.

Show your result

Hand in your reflection sheet.

Show your learning

Your reflection sheet. Everything else goes home.

Welcome to CIRC

I can find my way around CIRC and help set our trust and respect rules.

Find your seat

Find your seat.

Read the goal on the board.

Meet a partner

Tell a partner one thing you like to make.

How CIRC works

Look at our room map.

Find the trays, tools and the door line.

Trust and respect

With your table, write what trust looks like here.

Do the same for respect.

Quick build

Build the tallest tower you can with ten cards.

Test it. Improve it once.

Reset together

Return tools and reusable pieces.

Put unfinished work in the class tray.

Clear your table and floor.

Show your result

Say one rule you will keep in CIRC.

Show your learning

Your table's trust and respect ideas on the class chart.

Set Up Your Laptop

I can set up my school laptop so it is comfortable to use, and I can reach Teams, OneDrive and the right printer on my own.

Left and right click

Find the two bottom corners of the trackpad.

LEFT click picks something. It is the one you use most.

RIGHT click opens a menu of choices. You need it in one minute.

Left and right click (continued)

Practise: right-click on the desktop. A menu should appear. Press Escape to close it.

Best performance

RIGHT-click the battery icon at the bottom right.

Choose Power and battery settings.

Find the power mode boxes. There are TWO: On battery, and Plugged in.

Best performance (continued)

Set BOTH of them to Best performance.

Stay on this screen. You need it next.

Night light

Maximize this window so you can see the whole thing.

At the top there is a search box that says Find a setting. Type: night light.

Toggle Night light ON. Slide it ALL the way to the right.

Night light (continued)

Look at how orange that is. Now bring it back to 25 percent.

Close Settings.

Teams settings

Open Microsoft Teams and sign in with your school account.

Click the three dots next to your profile picture, then Settings.

You will see FOUR checkboxes. Turn them all OFF.

Teams settings (continued)

They are about auto-starting Teams, running in the background, and staying open when you close it.

Read the exact wording off your own screen and check your neighbour.

The blue cloud

Open OneDrive and look at the cloud icon.

BLUE cloud means you are signed in. You are done, help someone near you.

GREY cloud with a line through it means you are NOT signed in.

The blue cloud (continued)

If it is grey: LEFT-click it once and follow each prompt. Next, Next, Use this folder, Keep yes.

Hand up if it will not let you in.

Check each other

Check your partner: night light at 25, both power boxes on Best performance, blue cloud.

Only if there is time left, add the printer your teacher wrote on the board.

Respect the machine

These laptops are brand new. Close the lid gently with two hands.

Back on the cart, plugged in, cable not crushed.

Tell your teacher the one step that did not work for you.

Show your learning

Show the teacher your blue OneDrive cloud and Teams open.

Library launch: find it and ask a question

I can use call numbers or shelf signs to find a book and write a question it could answer.

Get ready

Sit with a partner.

Read the goal: find it and ask.

Tell your partner

Tell a partner about a book you liked.

How books are found

Watch how a call number points to a shelf.

Find it

Pick a topic you are curious about.

Use shelf signs or the catalog to find one book.

Write its call number.

Ask a question

Write one question the book could answer.

Check out the book if checkout runs today.

Reset together

Return books you are not keeping.

Push in chairs.

Show your result

Read your question to a partner.

Show your learning

One call number and one question the book could answer.

Fact card 1: Find and check a source

I can find three facts in a source I checked and write them in my own words.

Get ready

Get your fact sheet.

Read the goal: three checked facts.

Tell your partner

Tell a partner your topic.

Check a source

Look at two sources the teacher shows.

Which one would you trust? Why?

Find three facts

Find three facts in one checked source.

Write each fact in your own words.

Write the source

Write the title, the author or site, and the date.

Save your notes

Put your fact sheet in the class folder.

Return books and devices.

Show your result

Point to one fact you wrote in your own words.

Show your learning

Your fact sheet with three facts in your own words and the source.

Fact card 2: Design it and share

I can design a clear fact card with my three facts and source, then share it with my teacher.

Sign in

Get your device.

Open resources.svsd.net, then Canva.

Sign in with your school account.

Read a good card

Look at a fact card shown small.

What makes it easy to read?

Design your card

Start a blank design or a simple template.

Add your three facts and your source line.

Use two colors and one clear font.

Partner check

Trade screens. Can your partner read it from a step back?

Fix one thing.

Share with teacher

Choose Share.

Add your CIRC teacher's school account.

It is turned in when your teacher can open it.

Devices back

Close Canva.

Put your device back on the cart and plug it in.

Show your result

Tell a partner which fact is most surprising.

Show your learning

Share with your CIRC teacher's school account. It is turned in when your teacher can open it. Keep public and link sharing unchanged.

Mystery Materials

Test unknown material samples and select the best candidate for a stated design need.

Get ready

Place coded samples on the test mat.

Open the data table.

Sample safety

Use only approved tests.

Keep dropper water inside the tray.

Wipe each spill with a towel.

Sample safety (continued)

Taste or smell no sample.

Choose success rules

Read the design need.

Choose two things you can measure.

Write the success levels.

Run test one

Test one coded sample at a time.

Measure the result.

Record every code.

Run test two

Use the second approved property test.

Measure each sample again.

Record every result.

Choose a material

Compare all results with the success rules.

Choose one coded sample.

Reject one alternative with evidence.

Reset the samples

Return every coded sample.

Dry the test tray.

Exit check

Write the chosen code.

Cite two measured results.

Show your learning

Your chosen material with two test results and one tradeoff.

Hull Design

Shape a foil hull that stays stable while carrying the greatest tested payload.

Get ready

Place one foil sheet beside the tub.

Count the cargo pieces.

Water safety

Keep water inside the tub.

Wipe each spill with a towel.

Keep electronics outside the wet zone.

Plan the hull

Sketch a wide sealed hull.

Mark one likely weak point.

Shape the foil

Fold the edges above the water line.

Press the bottom flat.

Use no extra foil.

Float and load

Place the hull on the water.

Add one cargo piece at a time.

Record the last safe total.

Reshape the hull

Change one hull feature.

Repeat the load test.

Dry the station

Remove every cargo piece.

Dry the table.

Exit check

Write the better cargo total.

Show your learning

Your best cargo count and how hull shape helped.

Seed Travelers

Investigate seed-dispersal strategies and build a model seed that travels by wind, drop, float, or hitchhiking simulation.

Get ready

Choose one seed sample card.

Place the test sheet on your table.

Test safely

Keep water inside the tray.

Wipe each spill with a towel.

Keep every sample away from your mouth.

Sort the seeds

Study the seed shape.

Choose wind, drop, float, or hitchhiking.

Record your reason.

Build a model

Sketch one model seed.

Choose materials that match the travel method.

Build the model.

Run three trials

Use the matching test station.

Run one trial at a time.

Record each result.

Change one part

Change one shape or material.

Repeat the same test.

Reset the station

Dry the wet station.

Return every model piece.

Exit check

Circle the better result.

Name the useful feature.

Show your learning

Your model seed and its trial results.

Paper Circuit Signal

Build a safe paper circuit that uses one LED to communicate a clear signal or message.

Get ready

Place the circuit card on your table.

Keep the power switch off.

Power safety

Use only the switched battery holder.

Keep each battery inside the holder.

Switch off any warm holder.

Power safety (continued)

Report the warm holder to the teacher.

Plan the signal

Choose one clear signal word.

Draw one complete circuit path.

Mark which way the long light leg faces.

Teacher check

Hold up your paper plan.

Wait until the teacher says yes.

Do not take a battery box yet.

Build the circuit

Press copper tape along the path.

Make each corner smooth.

Place the LED in the marked direction.

Power the signal

Connect the holder leads to the marked points.

Turn on the switch.

Check the first place where the path breaks.

Finish the message

Test the light three times.

Add the signal word after success.

Power down

Turn off the switch.

Return the holder by count.

Exit check

Point to the complete circuit path.

Name one repair you made.

Show your learning

Your signal shown three times and the path traced.

Earthquake Platform

Build and improve a model structure that remains stable during a repeatable shake-table test.

Get ready

Place the platform on its boundary mat.

Check the height rule.

Platform safety

Keep fingers outside the moving platform.

Test only on the teacher signal.

Stop any falling part.

Plan for shaking

Sketch the structure.

Mark two likely weak joints.

Check the required footprint.

Build the structure

Build the base inside the footprint.

Reach the required height.

Check every connection.

Run the shake

Move hands behind the boundary.

Run the standard shake sequence.

Record movement or damage.

Strengthen one part

Improve one weak connection.

Repeat the same shake sequence.

Reset the platform

Remove loose pieces from the platform.

Sort reusable parts.

Exit check

Compare both damage scores.

Name the stronger feature.

Show your learning

Two damage scores and the change that helped.

Maze Coders

Write an efficient algorithm that moves a token through a grid maze without entering blocked cells.

Get ready

Place one maze card in the sleeve.

Put the token on start.

Token safety

Keep the token on the grid.

Pick up any dropped token at once.

Map the route

Mark the goal.

Mark each blocked cell.

Trace one possible route.

Write the code

Write one command per box.

Finish the full route before testing.

Partner test

Trade code with a partner.

Run each command exactly.

Mark the first error.

Debug the route

Change the first wrong command.

Remove one extra command if possible.

Run the route again.

Reset the maze

Return the token.

Erase the sleeve.

Exit check

Write the number of final commands.

Circle the command you changed.

Show your learning

Your final command count and the command that fixed the first bug.

Binary Beacons

Encode and decode a message using a documented two-state binary system.

Get ready

Choose two state cards.

Place the beacon at table level.

Light safety

Keep every light pointed at the table.

Turn off any light aimed near a face.

Use only classroom-safe messages.

Build the key

Assign a value to each state.

Write the value in each number spot.

Test one sample number.

Encode a message

Choose a short classroom message.

Convert each symbol with the key.

Place the states in order.

Send the states

Show one fixed position at a time.

Pause between symbols.

Let the partner record silently.

Check the message

Compare the partner answer with the original message.

Correct one unclear state.

Power down

Turn off every light.

Stack the state cards.

Exit check

Write one rule that prevented confusion.

Show your learning

Your key, coded message and decoded result.

Black Box Systems

Use controlled input-output tests to build and defend a model of a hidden system.

Get ready

Place the sealed system on its mat.

Open the data table.

Black box safety

Keep the box sealed.

Use only the approved test holes.

Move the box gently.

Choose test actions

List three safe test actions.

Change only one test action at a time.

Run controlled tests

Try the first test action.

Measure the result.

Record the matching action and result.

Run controlled tests (continued)

Repeat each action once.

Draw the system

Draw one possible inside system.

Connect each test action to a result.

Label the hidden parts.

Challenge the model

Choose one test that could show the drawing is wrong.

Run that test.

Revise the drawing if needed.

Reset the box

Remove every test piece.

Return the sealed system.

Exit check

Write one claim about the hidden system.

Cite one test result.

Show your learning

Your model, one supporting result and one question you still have.

Error-Proof Messages

Create a communication system that lets a receiver detect or correct a changed message.

Get ready

Choose one symbol key.

Place the channel envelope between teams.

Message safety

Use no personal information.

Use no secret information.

Use respectful classroom words.

Encode the message

Choose a short safe message.

Convert it with the symbol key.

Check every symbol once.

Add an error check

Choose repeat, number-sum check, even-or-odd check, or pattern.

Add the check to the message.

Write the checking rule.

Send through noise

Give the message to the change team.

Let the channel change one part.

Receive the changed message.

Detect the error

Apply the checking rule.

Mark the possible error.

Decode the message.

Reset the channel

Return every symbol card.

Recycle used message strips.

Exit check

Name the error check.

State whether it found the change.

Show your learning

The changed message and how the receiver caught it.

Biomimicry Grabber

Design a gripping mechanism inspired by how a plant or animal holds, hooks, wraps, or adheres.

Get ready

Choose one nature reference card.

Place lightweight objects on the test mat.

Grabber safety

Cut only on the table.

Keep fingers outside the cutting path.

Carry scissors closed.

Grabber safety (continued)

Test no body or live organism.

Copy a strategy

Name the nature trick.

Name its useful function.

Sketch a safe gripper.

Build the grabber

Cut only planned cardboard shapes.

Build the moving connection.

Add a soft place that can hold an object.

Test three objects

Grip the first approved object.

Record the result.

Repeat for two more objects.

Improve the grip

Change one nature-inspired part.

Repeat one failed object test.

Reset the grabber

Close the scissors.

Return every test object.

Exit check

Name the copied strategy.

Cite one test result.

Show your learning

Your grabber and the animal or plant idea it copies.

Rube Machine

Build a reliable chain reaction in which at least three distinct steps complete one simple task.

Get ready

Choose one final task card.

Place moving parts inside the table boundary.

Motion safety

Use only rolling tabletop parts.

Keep hands outside a moving path.

Stop any part that leaves the table.

Plan backward

Draw the final task first.

Add three earlier steps.

Mark where each part starts the next part.

Test each step

Build one step.

Test that step one time.

Fix the first part that does not work.

Connect the chain

Join the tested steps.

Try one full run.

Fix the first place where motion stops.

Connect the chain (continued)

Try one more full run if time remains.

Reset the parts

Stop every moving part.

Sort parts into labeled bins.

Remove tape scraps.

Exit check

Circle the hardest move from one part to the next.

Write the fix that worked.

Show your learning

Two complete runs and the handoff you fixed.

Check the AI answer

I can check claims in an AI-style answer against a trusted source and fix what is wrong.

Get ready

Get your AI answer card.

Read the goal: check, then fix.

Tell your partner

Tell a partner a time something online was wrong.

Watch a check

AI tools can sound sure and still be wrong.

Watch the teacher check one claim.

Check the claims

Read your AI answer card.

Find each claim in a book or checked source.

Mark it: matches, wrong, or cannot find.

Fix the answer

Rewrite one wrong claim correctly.

Add where you found it.

Reset together

Return books.

Put your card in the class folder.

Show your result

Read your fixed claim and its source.

Show your learning

Your checked card with one claim fixed and its source.

Stormwater Rescue

Model stormwater runoff and erosion, then test one intervention that slows, redirects, absorbs, or filters the flow.

Get ready

Place the sloped tray on the wet mat.

Measure the rain water.

Stormwater safety

Keep wet material inside the tray.

Wipe each spill with a towel.

Cut mesh only on the table.

Stormwater safety (continued)

Keep fingers outside the cutting path.

Taste no test material.

Run the first test

Pour the measured rain from the marked height.

Measure the water that leaves the tray.

Mark where soil moved.

Build a rescue

Choose one problem location.

Cut only the planned mesh piece.

Build one helper that slows or moves the water.

Repeat the rain

Pour the same water amount.

Measure the water that leaves the tray.

Mark where soil moved this time.

Improve the rescue

Change one part of the helper.

Predict the next result.

Clean the tray

One table at a time, carry wet soil to the return tub.

Place wet tools on the drying mat.

Dry the tray and table.

Clean the tray (continued)

Wash hands when the teacher calls your table.

Exit check

Compare both water amounts.

Name the helpful feature.

Show your learning

Baseline and revised results, and what your change did.

Code 1: Draw with a sequence

I can put code blocks in order to show pictures in the order I planned.

Open MakeCode

Get your laptop.

Open the MakeCode link on the board.

Tell your partner

Tell a partner a set of steps you follow every day.

Watch the blocks

Watch blocks snap together in order.

Watch the simulator run them.

Draw with code

Use show leds to make a picture.

Add a second picture after it.

Run it in the simulator.

Check the order

Does it show in the order you planned?

Swap two blocks and watch what changes.

Save and shut

Name your project with your initials.

Close the lid with two hands.

Put the laptop back and plug it in.

Show your result

Tell a partner what your pictures show, in order.

Show your learning

A program that shows your pictures in the order you planned.

Code 2: Loops make it move

I can use a loop to repeat pictures and make an animation.

Open MakeCode

Get your laptop.

Open the MakeCode link.

Recap

Tell a partner why order matters.

Watch a loop

Watch forever repeat the same steps.

Make it move

Make three pictures that change a little.

Put them in a forever loop.

Add a pause between them.

Debug

Find one thing that looks wrong.

Fix it and run again.

Show your animation to a partner.

Save and shut

Name your project with your initials.

Close the lid with two hands.

Put the laptop back and plug it in.

Tell what repeats

Tell a partner what repeats in your code.

Show your learning

An animation that repeats in a loop.

Code 3: If this, then that

I can make a program react differently when different buttons are pressed.

Open MakeCode

Get your laptop.

Open the MakeCode link.

Recap

Tell a partner what a loop does.

Watch a button

Watch on button A pressed change the picture.

Make it react

Show one picture when A is pressed.

Show another when B is pressed.

Test both buttons in the simulator.

Trade and test

Test a partner's program.

Tell them one bug or one win.

Save and shut

Name your project with your initials.

Close the lid with two hands.

Put the laptop back and plug it in.

Show your result

Say what happens when A is pressed.

Show your learning

A program that reacts differently to button A and button B.

Micro:bit Sensor Station

Program a classroom micro:bit or confirmed equivalent to detect a condition and produce a useful visual, sound, or motion alert.

Get ready

Place the micro:bit beside the computer.

Open the teacher starter file.

Connect only the approved cable.

Get ready (continued)

Wait until the computer shows the board.

Device safety

Use only approved USB power.

Connect no unknown device.

Handle the board by its edges.

Device safety (continued)

Disconnect power when the teacher says stop.

Plan the sensor

Choose one classroom condition.

Write what the board will measure.

Choose the number that turns the alert on.

Plan the sensor (continued)

Choose a clear alert.

Build the program

Place the measure block first.

Add the number check.

Add the alert block.

Build the program (continued)

Transfer the program to the approved device.

Test three values

Test one number below the turn-on number.

Test the turn-on number.

Test one number above it.

Test three values (continued)

Record each alert.

Tune the alert

Change the turn-on number or alert.

Repeat the three tests.

Disconnect safely

Stop the program.

Unplug the USB cable by its connector.

Return the board and cable to the tray.

Exit check

State the final turn-on number.

Name one limit of the data.

Show your learning

Three input tests and your sensor, threshold and alert.

Cardboard Arcade

Build a small tabletop cardboard skill game with one clear goal, a reliable mechanism, and a fast reset.

Get ready

Choose one tabletop game card.

Place tokens inside the play tray.

Game safety

Keep tokens inside the table boundary.

Stop any part that pinches.

Use no launcher.

Plan the game

Write the game goal in one sentence.

Sketch the token path.

Build the game

Build the smallest playable version.

Add one reliable cardboard connection.

Make a clear start point.

Play test

Ask a partner to play without help.

Watch the first confusing moment.

Record one unreliable part.

Improve the reset

Change the confusing part.

Reset the game twice.

Reset the arcade

Count every token.

Place the game on the storage shelf.

Exit check

State the goal.

Show how the game resets.

Show your learning

A playable game with a ten-second reset.

Outdoor Habitat Helper

Observe a local habitat need and design a safe, realistic helper for biodiversity, water, shelter, observation, or protective communication.

Get ready

Clip the site map to your board.

Choose one observation zone.

Walk out

Walk with your habitat team.

Habitat safety

Observe wildlife without touching it.

Leave nests, plants, droppings, or insects alone.

Cut model parts only on the table.

Habitat safety (continued)

Keep fingers outside the cutting path.

Install nothing outdoors.

Record the need

Record one habitat observation.

Name one need supported by that evidence.

Mark the location on the map.

Return inside

Walk back with your habitat team.

Place the map at your seat.

Plan a helper

Choose one helper type.

List one benefit.

List one risk.

Plan a helper (continued)

List one maintenance need.

Build a small model

Cut only approved model pieces.

Build the helper at table scale.

Add one clear safety label.

Pack the model

Close the scissors.

Place the model in the approval tray.

Exit check

Point to the evidence on your map.

Explain why teacher approval is still needed.

Show your learning

Your observation, helper, benefit and risk.

Outdoor Classroom Redesign

Survey the existing outdoor classroom and design one realistic improvement for learning, accessibility, organization, shade, seating, or habitat.

Get ready

Clip the site map to your board.

Choose one outdoor zone.

Walk out

Walk with your design team.

Stop at the assigned zone.

Measure safely

Keep the tape measure below your waist.

Leave plants untouched.

Stay on the approved surface.

Survey the zone

Record one feature that works.

Record one learning problem.

Measure the open space.

Draw the redesign

Name the user.

Draw the improvement on grid paper.

Label the main measurements.

Model one part

Build one key feature from paper.

Check its size against the drawing.

Pack the kit

Return each measuring tool.

Place the model in the class tray.

Return inside

Walk back with your design team.

Exit check

Write one reason the redesign helps.

Show your learning

Your redesign with the user need and the location marked.

Mirror Maze

Arrange reflective surfaces to direct a light path through checkpoints to a target.

Get ready

Place the light at the fixed source mark.

Keep the power off.

Light safety

Keep the light inside the table zone.

Point no light toward a face.

Point no reflection toward a window.

Light safety (continued)

Turn off the light before moving it.

Predict the path

Mark the source.

Mark each checkpoint.

Draw the expected light path.

Place the mirrors

Set one plastic mirror in its stand.

Match its angle to the drawing.

Repeat for each mirror.

Trace the light

Turn on the fixed light.

Trace the actual path on the grid.

Mark the first missed checkpoint.

Adjust the angles

Turn off the light.

Adjust one mirror.

Retest the full path.

Power down

Turn off the light.

Return mirrors to the padded tray.

Exit check

Draw the final path.

Circle the changed mirror.

Show your learning

The light path traced to the target and one angle fix.

Demo Day

Independently host, demonstrate, reset, and reflect on a polished project experience for an audience.

Get ready

Report to your assigned station.

Read your role card.

Station safety

Operate only approved demonstrations.

Keep every walkway clear.

Pause any station that cannot reset safely.

Check the station

Inspect the prototype.

Check the visitor boundary.

Practice the emergency pause.

Host the demo

Welcome the visitor.

State the design need.

Demonstrate the approved action.

Host the demo (continued)

Show the evidence.

Rotate the roles

Reset after each visitor.

Move to the next role card.

Answer questions with evidence.

Rotate the roles (continued)

Name a limit when needed.

Close the station

Thank the final visitor.

Complete the last safe reset.

Restore the room

Return all role cards.

Place every exhibit in storage position.

Exit check

Name one strong visitor moment.

Write one improvement for next time.

Show your learning

Your final reflection with one visitor observation.